Bitcoin Market Sentiment and Trader Performance Analysis

Objective: Analyze the relationship between Bitcoin market sentiment and trader performance using the Fear & Greed Index and Hyperliquid trading data.

Methodology

• Cleaned and transformed trading and sentiment datasets.
• Extracted and standardized dates.
• Merged both datasets on Date.
• Performed Exploratory Data Analysis (EDA) on profitability, trade size, trading activity, and Buy/Sell behavior.

Key Findings

1. Fear generated the highest trading activity with 61,837 trades.
2. Extreme Greed produced the highest average profit per trade (67.89 Closed PnL).
3. Fear delivered the highest total profit (3.36 million Closed PnL) due to increased trading volume.
4. Traders opened larger positions during Fear periods, with an average trade size of $7,816.
5. Sell activity increased during Greed and Extreme Greed, indicating profit-taking behavior.

Business Impact

Market sentiment significantly influences trader behavior, position sizing, and profitability. Integrating sentiment indicators into trading strategies can improve decision-making, risk management, and trading performance.

Tools Used: Python, Pandas, NumPy, Matplotlib

Skills Demonstrated: Data Cleaning, Data Merging, EDA, Visualization, Insight Generation, Business Analysis